Global Inference over Boundary-Head Candidates: A Research Programme

Every experiment this programme has run is catalogued chronologically in [[EXPERIMENT_CATALOG]], with its research line, outcome and real billed cost.

William Roe[1] (whr778@gmail.com), **Gemini**[2], and **Claude**[3] (noreply@anthropic.com)

[1] Project author and maintainer · [2] AI assistant (Google, Gemini) — diagnosis, architecture, experiments, and drafting · [3] AI assistant (Anthropic, Claude Opus 5) — design, implementation, and drafting

Programme map, revision of 2026-11-05. States the unifying thesis, the four papers it decomposes into, what is established (including the September–November 2026 document-level event and coreference line, `eb17` through `eb22g`), and what is still open. Numbers quoted here are summaries; each one's primary record is the working paper cited beside it.

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1. The thesis

GLiNER2's **boundary head** emits candidate scores per chunk — span boundaries, mention and pair logits, and contextual candidate states — and a **greedy per-chunk decode** then selects a record set from them. That decode is adequate for sparse, single-document extraction. The programme's claim is that it is not adequate for anything denser:

> **Dense document-level and beyond-document extraction require global structured inference over the boundary candidate scores, not the greedy per-chunk decode.**

The claim has two instantiations. They are the *same* top-K hypothesis inference at different scopes, over one shared `candidate_scores → JointProblem` contract:

scope	mechanism	what the beam ranges over	target task
within a document (combinatorial)	global decode under typed constraints	top-K constraint-consistent assignments	events (RAMS) **and** relations (Re-DocRED)
across documents (temporal)	EKF/MHT tracker — Kalman bank plus pruning	top-K hypotheses over time	evolving events in a document stream

Within a document, the two faces are one mechanism: in the beam an event is a *trigger node plus role edges* and a relation is a *plain edge*, so a single typed-constraint decode covers both. A win on only one face is a weaker but still reportable result, and the honest negative — that global decoding helps relations and not events, or the reverse — localises where greedy per-query decoding actually costs you.

Across documents the row is two questions, not one: **track** an evolving quantity, and **diarise** observations into the right stream. The distinction matters because the two halves fail independently, and the second can

silently destroy the first.

The boundary head is the shared substrate for both. It removed the span architecture's 19-instances-per-type cap, which had made either mechanism impossible to express.

2. What is established

The span head, not the encoder, was the bottleneck. Holding fresh heads fixed and swapping the encoder (mmBERT ↔ DeBERTa-v3) moves RAMS argument F1 barely at all (0.050 vs 0.042). Holding the encoder fixed and swapping fresh → IE-pretrained heads moves it roughly elevenfold (0.042 → 0.462). The deficit is head initialisation, not context window. *(Paper 0, §10.5.)*

Head warming has a measurable data threshold. Warming mmBERT's heads on a structure/argument corpus before RAMS fine-tuning does nothing at 10K, lifts arguments about 2.3× at 40K, and keeps climbing to 100K with no plateau. The knee is between 10K and 40K. *(Paper 0, §10.7.)*

Windowing, not the global decoder, recovers document-level arguments on a short-context model: 0.086 → 0.144 argument-strict F1 from matching the eval window to the trained window, against which the OneIE-style beam is neutral (0.144 → 0.137). Reported as a negative result for the decoder. *(Paper 0, §10.2.)*

Replay protects a warm start, and record extraction transfers without record supervision. Warm-starting the 137K joint base on real-plus-synthetic data with 30% exact replay held the original structure capability (0.1119 → 0.1060 on the base's own test set) while nearly tripling on the new distribution (0.0755 → 0.2179). The new corpora supplied *no* record-head supervision at all, so the gain is transfer from the span representations the record head reads its field fillers out of. *(`JOINT_IE_SCALING.md` §0b–0c.)*

An uncalibrated operating point cost more than any training intervention. *(2026-08-27.)* The stage-0 relevance gate ran its whole life at threshold 0.5. Its softmax is saturated and it needs 0.998 to sit at the stated recall bar; choosing the threshold on validation and scoring the blind test once moves overall accuracy 0.719 → 0.847 and the worst class 0.444 → 0.903, fixing 108 rows against 39 broken (exact McNemar $p = 1.1\times10^{-8}$). Two GPU fine-tuning runs and a four-way auxiliary label had been spent on that class; at each model's own validation threshold the rebuilt gate and its predecessor are indistinguishable (18/18, $p = 1.0000$). This is the §5 rule "quote a curve only with the operating point it was read at" recurring as a several-hundred-dollar lesson rather than a misdiagnosis. *(`GATES.md`; `EKF_MHT_DESIGN.md` §5.1.)*

Single-run variance on these metrics is ±0.02 or worse. A control re-run of a published RAMS recipe scored +0.023 above it from the re-run alone. Any curve claim needs at least two seeds per point before it is quoted, and several earlier readings of this programme's own curves did not meet that bar.

2f. Label negatives, and what a gold-derived menu hides (2026-09-17)

Established. Training and evaluation both built their label menu from each document's own gold, so no head except classification was ever shown a label it had to reject. Measured: **0 absent queries in 574** from

a real training batch, while `negative_query_ratio`, `abstention_loss` and `count_loss` — all live in every model this programme has trained — acted on exactly that empty set.

The consequence is a metric identity rather than a measurement: `event_type` has no span to get wrong, so a gold menu pins its precision at **1.0000** and `F1 = 2R/(1+R)` exactly, confirmed on 12 of 12 readings. Against a real taxonomy the same model scores precision **0.5521**, and against a menu of only ABSENT types it fires on **63 of 100 documents**.

Supplying absent labels is a genuine but bounded lever, and the trade is inherent rather than tunable: roughly **4x the argument-precision gain for a quarter of the recall gain**, with invented types down **363 → 274**. Two levers were tried against it — lowering the dose and halving the abstention weight — and neither moved it.

Method consequence, now a rule: *the menu is part of a metric's identity* (METRICS.md). A precision figure from a schema-driven task is uninterpretable without stating what the model was offered.

2g. Document-level event extraction with coreferent clusters (`eb17` → `eb22g`, 2026-09-15 to 2026-11-05)

Established. Between September and November 2026, the programme traced and solved four compounding bottlenecks that had held `event_argument` strict micro-F1 near `0.06–0.10` even when relaxed argument micro-F1 reached `0.50–0.59` (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`):

1. **Instance individuation by trigger span (`event_records: true`).** Prior to `eb17`, `event_records: true` was absent from every base config, so events were trained through the mention-structure path, which keys instances by *event type* (at most 1 per type per window) and pooled **64.2%** of multi-instance gold events (`EVENT_ARGUMENT_DIAGNOSIS.md`). Routing events through the record head in `natural` (`per_query`) mode individuates each event instance by its trigger anchor span (`record_instance_queries = 32`).
2. **Relational junction scoring (`record_link_mode: junction`).** Decomposing the record head's additive assignment logit `s_ijf = P_ij + R_fj` proved that role fit `R_fj` (identical across all triggers) accounted for 95% of row discrimination while pairwise compatibility `P_ij` sat at per-column AUC `0.576`, letting false triggers steal 52–74% of arguments (`JUNCTION_LAYER_SPEC.md`). Adding a role-gated bilinear + relative-geometry junction term trained with a contrastive per-column trigger softmax lifted per-column junction AUC to **`0.74–0.77`** and unlocked English argument binding (`p5link` `+0.035` strict micro-F1).
3. **Coreferent trigger and argument clusters (`record_coreferent_ownership`, `TriggerLink`, `EntityCorefLink`).** Training every coreferent trigger mention to own its event's arguments (`COREFERENT_OWNERSHIP_SPEC.md`), grouping coreferent and multi-occurrence argument spans as occurrence alternatives of a single filler (`RecordFieldTarget.values`), and adding a symmetric bilinear + 64-width MLP-distance `TriggerLink` scorer (pair AUC **`0.802–0.843`**, calibrated on validation by `coref_calibration.py`; `COREFERENT_LINK_SPEC.md`) together with `EntityCorefLink` (news pronoun AUC `0.685 -> 0.811`) enabled end-to-end document-level event cluster extraction (`event_cluster` and `event_cluster_argument`).

4. **Positive-gradient starvation in role assignment and span proposal (`eb22e` → `eb22f` → `eb22g`).** A paired-collate probe on `eb22e:checkpoint-epoch-1` (`3,568` gold role fillers) proved two simultaneous starvations: under scheduled injection (`0.956`), `97.1%` of gold fillers were in the candidate pool yet had median assignment probability `0.0155` because `_list_field_bce` averaged 1 positive with `K = 8` hard negatives in one denominator; with injection off (`0.0`), `61.7%` of gold fillers were missing from the pool or lost their trigger because `candidate_pair_loss` with `hard_negative_keep_all_when_absent: true` included all 768 candidates on absent queries (`~85–90%` of queries) in one global denominator, diluting positive pair gradients by 7,700× (`pair_loss` pinned at `0.005–0.011`) and leaving logit levels unanchored (`EB22C_SCORE_LEVEL_AUTOPSY.md`). Splitting positive and negative reductions in both `_list_field_bce` and `candidate_pair_loss` (`pair_split_reduction: true` , `hard_negative_keep_all_when_absent: false`), withholding unresolved gold roles, and eliminating double event dropout (`remove_events_prob: 0.0`) broke the `~0.25` CASIE proposal ceiling on 64-doc overfit (`overfit-events-eb22g` : CASIE `@0.5` trigger strict micro-F1 `0.3619 -> 0.5210` , argument `0.2506 -> 0.4319` ; CMNEE `@0.5` trigger `0.6375 -> 0.9394` , argument `0.5470 -> 0.8005`). On the full 263,949-document training mix (`eb22g`), Epoch 4 validation strict micro-F1 at swept threshold `@0.5` reaches `event_type` **`0.9524`** , `event_trigger` **`0.5656`** (recall **`0.7358`** , fair **`0.5916`**), `event_argument` **`0.3046`** (relaxed **`0.5223`** , fair **`0.5080`** , vs `eb22d` Epoch 2's `0.0627`), `event` **`0.6317`** (relaxed **`0.6818`**), `event_cluster` **`0.5681`** , `event_cluster_argument` **`0.2655`** (vs `eb22d` Epoch 2's `0.0579`), `entity` **`0.3689`** (relaxed `0.4455`), `relation` **`0.2145`** , `structure` **`0.2635`** , `classification` **`0.7166`** , `overall` **`0.4149`** , and `head_min` **`0.2145`** .

3. What is not established

The central question is now asked and answered — and the specified design was the wrong answer. *(Updated 2026-08-25.)* The design specified association as gate → Hungarian assignment → top-K hypotheses → track birth/death, and for a long time what shipped was hard assignment on an observable string key: no hypothesis enumeration, no deferred decision, no track birth or death. Every real-event failure recorded below is a failure of that placeholder.

What replaces it is **not** the specified design. A **global Viterbi decode over three states — own place, aggregate, reject** — improves every event we have at one setting: Helene 29.3 → 20.7 (−29.4%), Türkiye–Syria 11,581.5 → 10,695.5 (−7.6%), Aegean 2020 74.4 → 15.7 (−78.8%). Two properties carry it, and a hypothesis tree is neither: the decision must be **global**, because a greedy rule commits per observation and one large figure admitted early poisons a stream's scale for everything after; and it must be able to **reject**, because assignment headroom is measured at **zero** and the entire residual is the null hypothesis. Hungarian assignment optimises the half that is worth nothing. *(`EKF_MHT_DESIGN.md` §7.7.)*

On real news the tracker loses to a trivial baseline. On the Türkiye–Syria 2023 earthquake, pre-registered, `est_last_value` beats the EKF (0.208 vs 0.136), a 1999 death toll quoted in an article's history section is tracked as a 2023 figure, and one of the two affected countries is never recovered at all. Attribution, not filtering and not extraction, is the bottleneck. *(`EKF_MHT_DESIGN.md` §4.)*

The greedy-vs-beam comparison — the actual question of the combinatorial arm — HAS NOW BEEN RUN, and the thesis is not supported by it. *(2026-09-07, corrected 2026-09-08.)* One checkpoint (`eb16-rebuild-tr`), 18,786-record blind test, `boundary_head.decode_mode` as the only variable, strict micro F1:

> *Retracted — not at threshold 0.5; the card says 0.3. Current at the validation-selected 0.2: event_argument strict 0.0991 / relaxed 0.5884, event_trigger 0.5984, event_type 0.8650. See EVENT_ARGUMENT_DIAGNOSIS.md §4d.*

head	greedy	joint	Δ
entity	0.5695	0.5670	−0.0026
event_type	0.7545	0.7545	0.0000
event_argument	0.1178	0.1143	−0.0034
relation	0.1037	0.0994	−0.0043
structure	0.1640	0.1582	−0.0057
event_trigger	0.6051	0.5987	−0.0064
event	0.4081	0.4004	−0.0077

Every head is inside the ±0.02 floor, and the joint arm costs ~2.5× the wall clock (8.5 min against 22). Beam width is not the lever: a 16× sweep (4 / 16 / 64) moves structure by 0.0018, and *narrower* is marginally better, the opposite of a search-capacity story.

THE STRUCTURE ROW ABOVE IS A CORRECTION, and the size of it is the point. As first measured, structure read **0.1208 / 0.0754, a −0.0454 deficit** that was the one result outside the floor and the whole of the negative verdict. It was an artefact of a never-connected parameter: `field_dtypes_list` was declared, threaded and consumed correctly, and **no caller ever passed it**, so every `dtype: str` field compiled `ZERO_OR_MORE`. Cardinality selects the joint beam's utility (`logit − absent` against bare `logit`) and its exclusivity slot, so **the beam's entire scalar machinery — decision B of `JOINT_IE_DESIGN_RECORD` — had never engaged for a structure field.** Greedy looked almost right anyway because it re-derived the dtype at format time; joint had no such rescue. Connected, greedy gains +0.0432 and **joint gains +0.0828**, and 87% of the deficit disappears. The arm that was supposedly losing was the arm running with its machinery switched off.

The verdict does not change; its content does. "Joint decoding costs structure" is withdrawn. What stands is stronger and duller: **joint decoding matches greedy on all seven heads and costs 2.5× the wall clock.** A null at parity is a cleaner negative than a loss, because it cannot be explained away as a defect — and this one nearly was the other way round.

Three scopes on that negative, all load-bearing:

1. **It tests decoding-*with* the beam, not training-*for* it.** `JOINT_IE_DESIGN_RECORD` §7 (Phase B — beam in the loss) remains unrun and is the only version of the thesis this does not touch. It is also

now more interesting, not less: the beam has only ever been measured with its scalar constraints disconnected.

2. **A separate mechanism was tested first and also came back neutral.** `--global-decode` is the cross-window event *merge* (`assemble_events_global` operates on results already decoded), not inference over candidate scores. Neutral at the trained window, slightly negative when fragmented.
3. **Mechanism, not model — measured, on two independent lineages.** The pre-fix deficit reproduced on `joint-boundary-mmbert-137k-clean` (−0.0529) as well as on the control, so it never was a property of one checkpoint. A third pair (`casualty-multilingual-eb16tr`) is **unreadable** and was declared so by a rule written before the numbers existed: its greedy structure is 0.0070, a sixth of the floor, and its other heads sit at 0.0801 entity / 0.0209 event_type against the base's 0.5695 /

> *Retracted — not at threshold 0.5; the card says 0.3. Current at the validation-selected 0.2: event_argument strict 0.0991 / relaxed 0.5884, event_trigger 0.5984, event_type 0.8650. See EVENT_ARGUMENT_DIAGNOSIS.md §4d.* 0.7545 — catastrophic forgetting, not a decode result.

Two of this project's own instruments had to be repaired before that table could be read, and each hid the next. The scorer dropped list-shaped structure output in silence (48% of the apparent collapse). Then the first re-baseline measured a fix that never ran, because the schema carrier was dropped one layer above and **nothing in the log said whether the fix had executed** — so execution was inferred from the numbers it was meant to move. Runs now print `[records] compiled N field spec(s): X scalar ...` before scoring, and the third box was gated on that line rather than on hope.

Structure supervision now reaches the record head from every corpus on disk — this entry is a CORRECTION of the one it replaces. *(2026-09-08.)* A structure whose schema declares no `record_metadata` cannot be decoded by the record head, and the absence is valid and silent: no error, no warning, and the rows still count as supervision in every composition print. 60,948 rows across 13 models were in that state. The corpora were repaired in place, and the *producer* was fixed separately — `synthetic/validate.py` had gone on emitting structures without metadata, so every newly generated corpus reproduced the defect. **Audited across all of `data/`: zero corpora whose structures lack metadata.** `audit_corpora.py` carries a `STRUCT` check to keep it that way.

What remains is one level deeper and is a *decision*, not a defect: corpora declare only mode and anchor, never per-field **cardinality**, so a `dtype: str` field is still trained with list BCE rather than the scalar softmax over candidates plus `ABSENT`. Changing that moves every future model's structure numbers and needs an A/B first. *(`TODO.md`.)*

4. The four papers

Paper 0 — Foundation. *Schema-Driven Information Extraction Beyond the Sentence.* GLiNER2 plus the boundary head plus head-initialisation and multi-corpus training: the substrate the other three build on. Complete; no new experiments required. Design and results: `PAPER_0_FOUNDATION.md`.

Paper 1 — Real-time events across documents (temporal). The EKF/MHT line: censored measurement model, learned gate, text-to-observation normalisation, Viterbi three-state association (`own place` , `aggregate` , `reject`), held-out synthetic validation, and blind real events. Sequenced to resume immediately after `eb22g` completes, as EKF/MHT requires an upstream extractor that reliably binds arguments to specific event instances and clusters coreferent mentions within each document. Design and results: `EKF_MHT_DESIGN.md` .

Paper 2 — Constrained joint decoding & data scaling (combinatorial). Constrained joint selection wired to the boundary head (`+0.052` strict micro-F1 at `W=1` on Re-DocRED), the 10K–137K boundary data-scaling curve on repaired data, 30% exact replay, and the post-fix parity negative on multi-head beam decoding. Design and results: `JOINT_IE_SCALING.md` .

Paper 3 — Document-level event extraction with coreferent clusters. Instance individuation by trigger span (`event_records: true`), relational junction scoring and contrastive column loss (`record_link_mode: junction`), multi-mention trigger ownership and learned trigger/entity coreference (`TriggerLink` , `EntityCorefLink` , validation-calibrated cluster merging), occurrence-grouped coreferent argument compilation, and balanced positive-negative supervision (`pair_split_reduction` , split `_list_field_bce` , withheld unresolved roles) across the `eb17` → `eb22g` progression. Design and results: `PAPER_3_DOCUMENT_EVENTS_AND_COREF.md` .

Papers 1, 2, and 3 share the framing in §1 and cross-cite.

5. Method rules adopted after being learned the hard way

Verify splits before every run. A per-row random draw scattered copies of the same document across train, validation and test in 45 corpora. It is silent — nothing crashes and no metric looks wrong. Within-split overlap is now gated automatically before training, and Paper 0 §7.1 reports the audit and its effect on that paper's results rather than quietly restating the numbers.

Never compare across test sets. Two of this programme's retracted findings were cross-row comparisons on differently-composed blind tests. Support counts must match, or the comparison is not one.

Always report the trivial baseline beside the model. A pre-registered EKF prediction was scored a success by a range-normalised metric that was blind to contamination landing mid-range; `est_last_value` alongside it would have caught this immediately.

Quote a curve only with the operating point it was read at. A head whose decode threshold is never calibrated can read exactly zero while working correctly — which is how one head in this programme was misdiagnosed as broken across five measurements. It recurred in 2026-08-27 on the stage-0 gate, where the default threshold cost more accuracy than two fine-tuning runs recovered, and where a model comparison reported as significant ($p = 0.0091$) vanished entirely (18/18, $p = 1.0000$) once each model was read at its own threshold. **A comparison between two models at a shared arbitrary threshold measures the gap between two operating points, not between two models.**

A stored verdict is valid only for the model it was measured on. An ablation concluding that translation does not repair a multilingual gate was correct for the model it ran on — which reads the language and

over-admits — and exactly wrong for a later model that cannot read the language at all, where translation moved AUC 0.4733 → 0.8359. Re-run a cited verdict before applying it to a model that was not in it.

Graduation rule. A finding lives in its working paper until verified held-out, then graduates into the target paper. The working papers stay as the design and decision record; the papers carry only verified, reproducible claims — including the honest negatives that scope the contributions.

6. Source documents

The working papers are the primary record; this map is a summary of them.

document	role
`PAPER_0_FOUNDATION.md`	Paper 0 — substrate, head-init finding, data-integrity audit
`JOINT_IE_SCALING.md`	Paper 2 — joint decoding, 137K scaling curve, 30% replay, the three silent defects
`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`	Paper 3 — document-level events with coreferent clusters, relational junction scoring, and balanced positive-negative supervision (`eb17` → `eb22g`)
`JOINT_IE_DESIGN_RECORD.md`	Paper 2's build record — decisions, wiring map, cost model
`EVENT_LINE.md`	**line 3** — full event support on the record head: incumbent, risk register, and diagnostic trail
`JUNCTION_LAYER_SPEC.md`	gated bilinear + geometry trigger×argument junction layer and contrastive column loss
`COREFERENT_OWNERSHIP_SPEC.md`	multi-mention trigger ownership (`record_coreferent_ownership: true`)
`COREFERENT_LINK_SPEC.md`	learned `TriggerLink` and `EntityCorefLink` coreference scorers and validation-calibrated cluster merging
`EB22C_SCORE_LEVEL_AUTOPSY.md`	score-level swing autopsy, gradient-conflict measurement, and coref MLP distance discovery
`EVENT_ARGUMENT_DIAGNOSIS.md`	why mention-path event arguments fail — type pooling collapses 64.2% of multi-instance events
`PHASE_B_PLAN.md`	the beam **in the loss** — the only untested version of the thesis, with pre-registered bars and a throughput probe that gates the spend
`EKF_MHT_DESIGN.md`	Paper 1 — the filter, the real-event defeat, and the association half rebuilt as a global decode (§7.7)
`EKF_MHT_BUILD_RECORD.md`	Paper 1's build record — attachment points, generator spec, blind-test protocol
`GATES.md`	every gate in the pipeline, its type, and whether it is on
`THIRD_EVENT_AEGEAN2020.md`	the third event: pre-registration, and the prediction it falsified
`BOUNDARY_ARCHITECTURE.md`	how the boundary head works end to end, per task
`COUNTING_LAYER.md`	why the span 19-instance cap is a dead end
`BOUNDARY_DECODE_AND_EKF.md`	verified decode map and where global inference attaches
`RESEARCH_PROGRAM.md sec 1`	the beam ↔ filter origin analysis
`PROJECT_HISTORY.md`	chronological record of decisions, including those later overturned
`TODO.md`	open defects and next tests, with the evidence for each

document	role
`PIPELINES.md`	as-built vs as-designed maps, and the divergence table with M1–M5 status